

15.9 Solution: (a) For a screened Coulomb potential,

$$V(r) = \frac{zZe^2}{r} e^{-\alpha r},$$

the force on the particle is

$$\mathbf{F} = -\nabla V = zZe^2 \left(\alpha + \frac{1}{r} \right) \frac{e^{-\alpha r}}{r} \hat{r}.$$

In its componentwise form, by the Newton's second law, we have

$$m\ddot{x}(t) = zZe^2 \left(\alpha + \frac{1}{r} \right) \frac{e^{-\alpha r}}{r} \frac{x}{r}, \quad m\ddot{y}(t) = zZe^2 \left(\alpha + \frac{1}{r} \right) \frac{e^{-\alpha r}}{r} \frac{b}{r}.$$

Here, we have assumed that the particle follows a straight line, so that $y \equiv b$ and $r = \sqrt{x^2 + b^2}$. To apply the dipole approximation (see Problem 14.22), we need the Fourier transform of the accelerations,

$$\begin{aligned} \ddot{x}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \ddot{x}(t) e^{i\omega t} dt = \frac{zZe^2}{\sqrt{2\pi}m} \int_{-\infty}^{+\infty} \left(\alpha + \frac{1}{r} \right) \frac{e^{-\alpha r}}{r} \frac{x}{r} e^{i\omega t} dt \\ &= \frac{\alpha zZe^2}{\sqrt{2\pi}m} \int_{-\infty}^{+\infty} \left(\frac{1}{x^2 + b^2} + \frac{1}{\alpha(x^2 + b^2)^{3/2}} \right) x \exp(-\alpha\sqrt{x^2 + b^2}) e^{i\omega t} dt. \end{aligned}$$

Since $x = vt$ for a straight path, and also notice the parity of the integrand, we can write the Fourier transform of the acceleration in the x -direction as

$$\ddot{x}(\omega) = \frac{i\alpha zZe^2}{\sqrt{2\pi}mv} \int_{-\infty}^{+\infty} \left(\frac{1}{x^2 + b^2} + \frac{1}{\alpha(x^2 + b^2)^{3/2}} \right) x \exp(-\alpha\sqrt{x^2 + b^2}) \sin\left(\frac{\omega}{v}x\right) dx.$$

Similarly,

$$\ddot{y}(\omega) = \frac{\alpha bzZe^2}{\sqrt{2\pi}mv} \int_{-\infty}^{+\infty} \left(\frac{1}{x^2 + b^2} + \frac{1}{\alpha(x^2 + b^2)^{3/2}} \right) \exp(-\alpha\sqrt{x^2 + b^2}) \cos\left(\frac{\omega}{v}x\right) dx.$$

Using the integration identities (see Gradshteyn and Ryzhik, 8th ed., 3.914.5 and 3.914.10, pg. 522),

$$\begin{aligned} \int_0^{+\infty} \left(\frac{1}{x^2 + b^2} + \frac{1}{\alpha(x^2 + b^2)^{3/2}} \right) x \exp(-\alpha\sqrt{x^2 + b^2}) \sin(\gamma x) dx &= \frac{\gamma}{\alpha} K_0(b\sqrt{\alpha^2 + \gamma^2}), \\ \int_0^{+\infty} \left(\frac{1}{x^2 + b^2} + \frac{1}{\alpha(x^2 + b^2)^{3/2}} \right) \exp(-\alpha\sqrt{x^2 + b^2}) \cos(\gamma x) dx &= \frac{\sqrt{\alpha^2 + \gamma^2}}{\alpha b} K_1(b\sqrt{\alpha^2 + \gamma^2}), \end{aligned}$$

the above Fourier transforms can be *exactly* evaluated, with $\gamma \rightarrow \omega/v$,

$$\ddot{x}(\omega) = \frac{2i\omega zZe^2}{\sqrt{2\pi}mv^2} K_0 \left(b\sqrt{\alpha^2 + \frac{\omega^2}{v^2}} \right), \quad \ddot{y}(\omega) = \frac{2zZe^2}{\sqrt{2\pi}mv} \sqrt{\alpha^2 + \frac{\omega^2}{v^2}} K_1 \left(b\sqrt{\alpha^2 + \frac{\omega^2}{v^2}} \right).$$

From the dipole approximation,

$$\frac{dI}{d\omega}(\omega, b) = \frac{4}{3c^3} \left| \ddot{\mathbf{d}}(\omega) \right|^2,$$

where $\ddot{\mathbf{d}}(\omega) = ze(\ddot{x}(\omega), \ddot{y}(\omega))$, we know

$$\frac{dI}{d\omega}(\omega, b) = \frac{8}{3\pi c^3} \left(\frac{Zz^2 e^3}{mv} \right)^2 \left[\frac{\omega^2}{v^2} K_0 \left(b \sqrt{\alpha^2 + \frac{\omega^2}{v^2}} \right)^2 + \left(\alpha^2 + \frac{\omega^2}{v^2} \right) K_1 \left(b \sqrt{\alpha^2 + \frac{\omega^2}{v^2}} \right)^2 \right].$$

For $\omega v/b \gg 1$, $b\sqrt{\alpha^2 + \omega^2/v^2} \gg 1$. By the asymptotic behavior of the modified Bessel function, $K_\nu(x) \sim e^{-x}$ as $x \rightarrow +\infty$, we can see that the radiated energy is almost negligible. On the other, when $\omega \ll v/b$, we can safely drop ω/v in the above expression and will obtain

$$\frac{dI}{d\omega}(\omega, b) = \frac{8}{3\pi c^3} \left(\frac{Zz^2 e^3}{mv} \right)^2 \alpha^2 K_1(\alpha b)^2 = \frac{8}{3\pi} \frac{Z^2 e^2}{c} \left(\frac{z^2 e^2}{mc^2} \right)^2 \left(\frac{c}{v} \right)^2 \alpha^2 K_1(\alpha b)^2.$$

(b) The radiation cross section is

$$\begin{aligned} \frac{d\chi}{d\omega} &= \int_{b_{\min}}^{b_{\max}} \frac{dI}{d\omega}(\omega, b) \cdot 2\pi b db = \frac{16}{3} \frac{Z^2 e^2}{c} \left(\frac{z^2 e^2}{mc^2} \right)^2 \left(\frac{c}{v} \right)^2 \int_{b_{\min}}^{b_{\max}} \alpha^2 K_1(\alpha b)^2 b db \\ &= \frac{16}{3} \frac{Z^2 e^2}{c} \left(\frac{z^2 e^2}{mc^2} \right)^2 \left(\frac{c}{v} \right)^2 \int_{\alpha b_{\min}}^{\alpha b_{\max}} x K_1(x)^2 dx. \end{aligned}$$

Since

$$\begin{aligned} \int x K_1(x)^2 dx &= \frac{1}{2} x^2 [K_1(x)^2 - K_0(x) K_2(x)] + C \\ &= \frac{1}{2} x^2 \left[K_1(x)^2 - K_0(x) \left(K_0(x) + \frac{2}{x} K_1(x) \right) \right] + C \\ &= -\frac{1}{2} x^2 \left[K_0(x)^2 - K_1(x)^2 + \frac{2}{x} K_0(x) K_1(x) \right] + C, \end{aligned}$$

then

$$\frac{d\chi}{d\omega} = -\frac{16}{3} \frac{Z^2 e^2}{c} \left(\frac{z^2 e^2}{mc^2} \right)^2 \left(\frac{c}{v} \right)^2 \left\{ \frac{1}{2} x^2 \left[K_0(x)^2 - K_1(x)^2 + \frac{2}{x} K_0(x) K_1(x) \right] \right\} \Big|_{\alpha b_{\min}}^{\alpha b_{\max}}.$$